JULIO LANDA

PhD Student in Renewable Energy
Mechatronics Engineer

EDUCATION

PhD in Renewable Energy

Instituto de Energías Renovables, UNAM · 2026-present
PhD Supervisor: Dr. Camilo Arancibia
Research: Closed-loop control of solar heliostats using computer vision and complementary sensing methods.

Master's in Engineering - Energy Area

Instituto de Energías Renovables, UNAM · 2022-2025
Morelos, Mexico.

Bachelor of Mechatronics Engineering

Universidad Tecnológica de Emiliano Zapata · 2018-2020
Emiliano Zapata, Morelos, Mexico.

Higher University Technician in Mechatronics, Automation Area

Universidad Tecnológica de Emiliano Zapata · 2016-2018
Emiliano Zapata, Morelos, Mexico.

CURRENT RESEARCH

Sensor-based feedback for heliostat control

IER-UNAM · 2026-present

Development of sensor-based feedback methods for closed-loop heliostat positioning, with particular emphasis on computer vision. Experimental work with the heliostat, solar furnace, and control system at IER-UNAM.

PUBLICATIONS

IoT smartwatch based on open technologies for the collection of thermal comfort data

Landa, J., Barrios, G., & Huelsz, G. (2025).
HardwareX, 22, e00633.
DOI: 10.1016/j.ohx.2025.e00633

THESES

IoT smartwatch based on open technologies for the collection of thermal comfort data

Master's thesis · 2025
Available at: https://altamarmx.github.io/reloj_inteligente/
Code repository: GitHub · GPL v3.0

PROFESSIONAL EXPERIENCE

NC School Cuernavaca

Lead Physics and Mathematics Teacher · 2024-2025

Lead teacher for physics and mathematics courses.

IER, UNAM

Teaching Assistant · 2023-present

Currently support the Solar Thermal Energy course, particularly solar geometry. Previously assisted with assignments, exams and project assessment for Introduction to Bioclimatic Design, and served as teaching assistant for Linear Algebra.

IoT Developer Engineer · 2020 / 2024-present

Develop low-cost devices using open-source hardware and software, integrated with IoT platforms for the instrumentation project of the new Renewable Energy Engineering building at IER, UNAM.

JL

CONTACT

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JulioLanda4

Website

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Google Scholar

Morelos, Mexico

LANGUAGES

- Spanish: native
- English: Professional working proficiency

RESEARCH INTERESTS

Control Systems · Mechatronics · Computer Vision · Solar Energy · Instrumentation · Embedded Systems

SKILLS

- **Instrumentation:** instrument design and calibration.
- **Programming:** Python, C/C++, Arduino and LabVIEW.
- **Control:** feedback control, automation, sensor-based control, and experimental control systems.
- **Mechanical design:** CAD modeling with SolidWorks and FreeCAD.
- **Data analysis:** Python, NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Seaborn and Plotly.
- **Artificial intelligence:** machine learning, neural networks and genetic algorithms.
- **Embedded systems and IoT:** ESP32/Arduino development and platform integration.
- **Electronics:** circuit design and automation projects.

CNEER

General Coordinator · 2026

General Coordinator of the National Congress of Renewable Energy Students (CNEER) 2026.

CNEER

Workshop Coordinator · 2023

Planned, organized and coordinated workshops for the 10th edition of CNEER, the National Congress of Renewable Energy Students.

DMC Ingenieria

Electrical Engineer · 2022

Provided maintenance to electric motors and control panels as a contractor for Cementos Moctezuma.

Bombas y Servicios Corbala

Electrical Engineer · 2021

Maintained and installed deep-well water pumps, and designed and installed control systems and electrical panels.

Alucaps

Maintenance Technician · 2020

Applied maintenance plans to company machinery, updated and programmed production-line control systems using PLCs, and assembled electrical panels.

CERTIFICATIONS AND ADDITIONAL TRAINING

Certifications

- **Certified SolidWorks Professional (CSWP)** · 2019.
- **Google IT Automation with Python** · 2022.
- **LabVIEW Associate Developer** · 2019.

Courses

- **Macrotraining in Artificial Intelligence (MeIA)** · 2023.
- **Python for Engineering (UNAM, Coursera)** · 2023.

HONORS AND ACADEMIC ENGAGEMENT

- **Recognition** · 2026. For serving as General Coordinator of the National Congress of Renewable Energy Students (CNEER) 2026.
- **Recognition** · 2026. Specialized Workshop “Data Science and Methodology for Climate Analysis”, delivered on May 19, 2026 at the International Congress of Technology and Applied Science, Cuernavaca, Morelos, Mexico.
- **Recognition** · 2026. Participation as part of the organizing committee for Hack-ODS Hackathon 2026, organized by the Institute of Renewable Energy and the Institute of Engineering at UNAM, from January 12 to April 24, 2026.
- **Honorable Mention** · 2025. XXXV CUAM-ACMor Research Congress, Physics-Mathematics category, project “Green Motion: How Do You Move?”.
- **Recognition as speaker** · 2024. Sustainable Buildings Symposium: Simulations and Digitalization for Decarbonization.
- **Honorable Mention** · 2023. XXXIII CUAM-ACMor Research Congress, Physics-Mathematics category, project “Ecological Greenhouse from Plastic Waste”.
- **1st place** · 2019. 3rd Mathematics Olympiad, Universidad Tecnológica de Emiliano Zapata.